Draft Standard for
Local and metropolitan area networks —

Frame Replication and Elimination for Reliability

Corrigendum 1: Technical and Editorial Corrections

Prepared by the Maintenance Task Group of IEEE 802.1

LAN MAN Standards Committee of the
IEEE Computer Society

Abstract: This corrigendum to IEEE Std 802.1CB™-2017 provides technical and editorial corrections.

Keywords: TSN, Time-Sensitive Networking, Redundancy, Bridging, Bridges, Frame Replication, Frame Elimination, Stream identification, Extended Stream identification, Bridged Local Area Networks, IEEE 802®, IEEE 802.1Q™, IEEE 802.1CB™, IEEE 802.1CBcv™, IEEE 802.1CBdb™, local area networks (LANs), MAC Bridges, Virtual Bridged Local Area Networks (virtual LANs), amendment.

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[5] Available at: https://standards.ieee.org/about/sasb/patcom/materials.html.

Participants

The following is a list of participants in the IEEE 802.1 Working Group.

Glenn Parsons, *Chair*
Jessy Rouyer, *Vice-Chair*
Mark Hantel, *Chair, Maintenance Task Group*
Christophe Mangin, *Editor*

The following members of the individual balloting committee voted on this standard. Balloters may have voted for approval, disapproval, or abstention.

When the IEEE-SA Standards Board approved this standard on XX Month 20XX, it had the following membership:

Name, *Chair*
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Name, *Past Chair*

Introduction

> This introduction is not part of IEEE P 802.1CB-2017/Cor1-20XX, IEEE Draft Standard for Local and metropolitan area networks – Frame Replication and Elimination for Reliability – Corrigendum 1.

The first edition of IEEE Std 802.1CB was published in 2017. A first amendment IEEE Std 802.1CBcv-2021 added the YANG data model that allows configuration and state reporting for all managed objects of the base standard. A second amendment IEEE Std 802.1CBdb-2021 added the mask-and-match Stream identification function and provided technical and editorial corrections.

This corrigendum IEEE 802.1CB-2017/Cor1-2025 implements the changes necessitated by various maintenance items that have been notified to the IEEE 802.1 Working Group. The maintenance items addressed by this corrigendum are: 315, 338, 344 and 364.

Contents

List of figures

Draft Standard for Local and metropolitan area networks —

Frame Replication and Elimination for Reliability

Corrigendum 1: Technical and Editorial Corrections

NOTE—The editing instructions contained in this corrigendum define how to merge the material contained here into the base document and its other amendments to form the new comprehensive standard.

The editing instructions are shown in ***bold italic***. Four editing instructions are used: ***change***, ***delete***, ***insert***, and ***replace***. ***Change*** is used to make corrections in existing text or tables. The editing instruction specifies the location of the change and describes what is being changed either by using ~~strikethrough~~ (to remove old material) and underscore (to add new material). ***Delete*** removes existing material. ***Insert*** adds new material without disturbing the existing material. **Insertion** may require renumbering. If so, renumbering instructions are given in the editing instruction. ***Replace*** is used to make changes in figures or equations by removing the existing figure or equation and replacing it with a new one. Editing instructions, change markings, and this NOTE will not be carried over into future editions because the changes will be incorporated into the base standard.[6]

[6]Notes in text, tables, and figures are given for information only, and do not contain requirements needed to implement the standard.

6. Stream identification

6.2 Stream identification function

Insert the following paragraph at the end of the clause, before the notes, in IEEE Std 802.1CB-2017:

The null-value is a value known from the management system as identifying the stream_handle assigned to packets that belong to no known Stream.

6.7 IP Stream identification

Insert the following text after the second paragraph:

Practically, the IP Stream identification function requires locating the beginning of the IP packet encapsulated in the frame, preceded by an IP EtherType field, e.g., 0x0800 or 0x88DD. The location of the IP EtherType field within the frame is dependent on the combination of tag(s) following the addresses in a frame. The method used to determine this location is beyond the scope of this standard.

9. Stream Identification Management

9.1 Stream identity table

9.1.1 tsnStreamIdEntry

9.1.1.1 tsnStreamIdHandle

Insert the following note at the end of the subclause:

NOTE—It is also possible to assign a stream_handle of 0 to a packet that was identified as belonging to a known stream (i.e., a specific tsnStreamIdEntry).

12. YANG Data Model

12.1 YANG Framework

Change the text in the lettered-list as shown:

This clause has been developed according to the YANG guidelines published in IETF RFC 6087 [B5] as applicable to IEEE standards.

The YANG framework applies hierarchy in the following areas:

a) The uniform resource name (URN), as specified in IEEE Std 802d [B2a]. ~~The structure of the URN is such that ieee is the root (i.e., name-space identifier), followed by the standard, then the working group developing the standard.~~
b) The YANG objects form a hierarchy of configuration and operational data structures that define the YANG model. These hierarchical relationships are described in 12.2.

12.2 IEEE Std 802.1CB YANG model

12.2.2 Frame Replication and Elimination for Reliability model

Replace Figure 12-5 with the following figure:

sequence-generation (10.3)		
uint32	index	// r-w
handle-ref	* stream	// r-w
boolean	direction-out-facing	// r-w
boolean	reset	// r-w

sequence-recovery (10.4)		
uint32	index	// r-w
handle-ref	* stream	// r-w
if-ref	* port	// r-w
boolean	direction-out-facing	// r-w
boolean	reset	// r-w
sequence-recovery-algorithm	algorithm	// r-w
sequence-history-length	history-length	// r-w
uint32	reset-timeout	// r-w
uint32	invalid-sequence-value	// r
boolean	take-no-sequence	// r-w
boolean	individual-recovery	// r-w
boolean	latent-error-detection	// r-w
int32	latent-error-difference	// r-w
uint32	latent-error-period	// r-w
uint16	latent-error-paths	// r-w
uint32	latent-error-reset-period	// r-w

sequence-identification (10.5)		
handle-ref	* stream	// r-w
if-ref	port	// r-w
boolean	direction-out-facing	// r-w
boolean	active	// r-w
enum	encapsulation	// r-w
int8	path-id-lan-id	// r-w

stream-split (10.6)		
uint32	index	// r-w
if-ref	port	// r-w
boolean	direction-out-facing	// r-w
leaf-ref	* input-id	// r-w
leaf-ref	* output-id	// r-w

sequence-autoconfiguration (10.7.1)		
uint32	index	// r-w
enum	sequence-encapsulation	// r-w
if-ref	* receive-port	// r-w
enum	tagged	// r-w
dot1qtypes:vlanid	* vlan	// r-w
if-ref	* recovery-port	// r-w
uint64	destruction-interval	// r-w
uint64	reset-interval	// r-w
sequence-recovery-algorithm	algorithm	// r-w
sequence-history-length	history-length	// r-w
boolean	create-individual	// r-w
boolean	create-recovery	// r-w
boolean	latent-error-detection	// r-w
int32	latent-error-difference	// r-w
uint32	latent-error-period	// r-w
uint32	latent-error-reset-period	// r-w

output-autoconfiguration (10.7.2)		
uint32	index	// r-w
if-ref	* port	// r-w
enum	encapsulation	// r-w
int8	lan-path-id	// r-w

Figure 12-5—FRER model

12.6 Definition of 802.1CB YANG modules[7] [8]

Change clause header as shown below:

12.6 Definition of IEEE Std 802.1CB YANG modules

12.6.1.2 YANG data scheme definition for ieee802-dot1cb-frer YANG module

Change the text as shown:

```
module: ieee802-dot1cb-frer
  +--rw frer
     +--rw sequence-generation* [index]
|  +--rw index                   uint32
     |  +--rw stream*                 -> /dot1cb-sid:stream-identity/handle
     |  +--rw direction-out-facing?   dot1cb-sid-types:direction
     |  +--rw reset?                  boolean
     +--rw sequence-recovery* [index]
     |  +--rw index                                uint32
     |  +--rw stream*
     |  |       -> /dot1cb-sid:stream-identity/handle
     |  +--rw port*                                if:interface-ref
     |  +--rw direction-out-facing?                dot1cb-sid-types:direction
     |  +--rw reset?                               boolean
     |  +--rw algorithm
     |  |  +--rw (algorithm)?
     |  |     +--:(vector)
     |  |     |  +--rw vector!
     |  |     |     +--ro type-number?   dot1cb-frer-types:seq-rcvy-algorithm
     |  |     |     +--ro oui-cid?       string
     |  |     +--:(match)
     |  |     |  +--rw match!
     |  |     |     +--ro type-number?   dot1cb-frer-types:seq-rcvy-algorithm
     |  |     |     +--ro oui-cid?       string
     |  |     +--:(organization-specific)
     |  |        +--rw organization-specific!
     |  |           +--rw type-number?   int32
     |  |           +--rw oui-cid?       string
     |  +--rw history-length?                      sequence-history-length
     |  +--rw reset-timeout?                       uint32
     |  +--ro invalid-sequence-value?              uint32
     |  +--rw take-no-sequence?                    boolean
     |  +--rw individual-recovery?                 boolean
     |  +--rw latent-error-detection?              boolean
     |  +--rw latent-error-detection-parameters
     |     +--rw difference?     int32
     |     +--rw period?         uint32
     |     +--rw paths?          uint16
     |     +--rw reset-period?   uint32
     +--rw sequence-identification* [port direction-out-facing]
     |  +--rw stream*                 -> /dot1cb-sid:stream-identity/handle
     |  +--rw port                    if:interface-ref
```

[7]Copyright release for YANG modules: Users of this standard may freely reproduce the YANG modules contained in this subclause so that they can be used for their intended purpose.

[8]An ASCII version of the YANG modules are attached to the PDF version of this standard, and can be obtained by Web browser from the IEEE 802.1 Website at https://1.ieee802.org/yang-modules/.

```
   |  +--rw direction-out-facing    dot1cb-sid-types:direction
   |  +--rw active?                 boolean
   |  +--rw encapsulation
   |  |  +--rw (encapsulation)?
   |  |     +--:(r-tag)
   |  |     |  +--rw r-tag!
   |  |     |     +--ro type-number?   dot1cb-frer-types:seq-encaps-method
   |  |     |     +--ro oui-cid?       string
   |  |     +--:(hsr-sequence-tag)
   |  |     |  +--rw hsr-sequence-tag!
   |  |     |     +--ro type-number?   dot1cb-frer-types:seq-encaps-method
   |  |     |     +--ro oui-cid?       string
   |  |     +--:(prp-sequence-tag)
   |  |     |  +--rw prp-sequence-tag!
   |  |     |     +--ro type-number?   dot1cb-frer-types:seq-encaps-method
   |  |     |     +--ro oui-cid?       string
   |  |     +--:(organization-specific)
   |  |        +--rw organization-specific!
   |  |           +--rw type-number?   int32
   |  |           +--rw oui-cid?       string
   |  +--rw path-id-lan-id?         lan-path-id
   +--rw stream-split* [~~port direction-out-facing~~index]
   |  +--rw index                   uint32
   |  +--rw port                    if:interface-ref
   |  +--rw direction-out-facing    dot1cb-sid-types:direction
   |  +--rw input-id*               -> /dot1cb-sid:stream-identity/handle
   |  +--rw output-id*              -> /dot1cb-sid:stream-identity/handle
   +--rw autoconfiguration {auto-configuration}?
      +--rw sequence* [index]
      |  +--rw index                           uint32
      |  +--rw sequence-encapsulation
      |  |  +--rw (encapsulation)?
      |  |     +--:(r-tag)
      |  |     |  +--rw r-tag!
      |  |     |     +--ro type-number?
      |  |     |     |       dot1cb-frer-types:seq-encaps-method
      |  |     |     +--ro oui-cid?       string
      |  |     +--:(hsr-sequence-tag)
      |  |     |  +--rw hsr-sequence-tag!
      |  |     |     +--ro type-number?
      |  |     |     |       dot1cb-frer-types:seq-encaps-method
      |  |     |     +--ro oui-cid?       string
      |  |     +--:(prp-sequence-tag)
      |  |     |  +--rw prp-sequence-tag!
      |  |     |     +--ro type-number?
      |  |     |     |       dot1cb-frer-types:seq-encaps-method
      |  |     |     +--ro oui-cid?       string
      |  |     +--:(organization-specific)
      |  |        +--rw organization-specific!
      |  |           +--rw type-number?   int32
      |  |           +--rw oui-cid?       string
      |  +--rw receive-port*             if:interface-ref
      |  +--rw tagged?                   enumeration
      |  +--rw vlan*                     dot1qtypes:vlanid
      |  +--rw recovery-port*            if:interface-ref
      |  +--rw destruction-interval?     uint64
      |  +--rw reset-interval?           uint64
      |  +--rw algorithm
      |  |  +--rw (algorithm)?
```

```
      |  |     +--:(vector)
      |  |     |  +--rw vector!
      |  |     |     +--ro type-number?
      |  |     |     |       dot1cb-frer-types:seq-rcvy-algorithm
      |  |     |     +--ro oui-cid?       string
      |  |     +--:(match)
      |  |     |  +--rw match!
      |  |     |     +--ro type-number?
      |  |     |     |       dot1cb-frer-types:seq-rcvy-algorithm
      |  |     |     +--ro oui-cid?       string
      |  |     +--:(organization-specific)
      |  |        +--rw organization-specific!
      |  |           +--rw type-number?   int32
      |  |           +--rw oui-cid?       string
      |  +--rw history-length?              sequence-history-length
      |  +--rw create-individual?           boolean
      |  +--rw create-recovery?             boolean
      |  +--rw latent-error-detection?      boolean
      |  +--rw latent-error-difference?     int32
      |  +--rw latent-error-period?         uint32
      |  +--rw latent-error-reset-period?   uint32
      +--rw output* [index]
         +--rw index            uint32
         +--rw port*            if:interface-ref
         +--rw encapsulation
         |  +--rw (encapsulation)?
         |     +--:(r-tag)
         |     |  +--rw r-tag!
         |     |     +--ro type-number?
         |     |     |       dot1cb-frer-types:seq-encaps-method
         |     |     +--ro oui-cid?       string
         |     +--:(hsr-sequence-tag)
         |     |  +--rw hsr-sequence-tag!
         |     |     +--ro type-number?
         |     |     |       dot1cb-frer-types:seq-encaps-method
         |     |     +--ro oui-cid?       string
         |     +--:(prp-sequence-tag)
         |     |  +--rw prp-sequence-tag!
         |     |     +--ro type-number?
         |     |     |       dot1cb-frer-types:seq-encaps-method
         |     |     +--ro oui-cid?       string
         |     +--:(organization-specific)
         |        +--rw organization-specific!
         |           +--rw type-number?   int32
         |           +--rw oui-cid?       string
         +--rw lan-path-id?     lan-path-id

augment /dot1cb-sid:stream-identity:
  +--ro auto-configured?   boolean {auto-configuration}?
  +--rw lan-path-id?       lan-path-id {auto-configuration}?
augment /if:interfaces/if:interface/if:statistics:
  +--ro frer
     +--ro per-port-counters
     |  +--ro rx-passed-pkts?        uint64
     |  +--ro rx-discarded-pkts?     uint64
     |  +--ro encode-errored-pkts?   uint64
     +--ro per-port-per-stream-counters* [direction-out-facing handle]
        +--ro direction-out-facing      dot1cb-sid-types:direction
        +--ro handle
```

```
         |       -> /dot1cb-sid:stream-identity/handle
         +--ro generation-reset?         uint64
         +--ro rx-out-of-order-pkts?     uint64
         +--ro rx-rogue-pkts?            uint64
         +--ro rx-passed-pkts?           uint64
         +--ro rx-discarded-pkts?        uint64
         +--ro rx-lost-pkts?             uint64
         +--ro rx-tagless-pkts?          uint64
         +--ro rx-resets?                uint64
         +--ro rx-latent-error-resets?   uint64
         +--ro encode-errored-pkts?      uint64
```

12.6.2 YANG data module definitions

12.6.2.4 Definition for the ieee802-dot1cb-frer YANG module

Change the text as shown:

```
module ieee802-dot1cb-frer {
  yang-version "1.1";
  namespace urn:ieee:std:802.1Q:yang:ieee802-dot1cb-frer;
  prefix dot1cb-frer;
  import ieee802-dot1cb-stream-identification {
    prefix dot1cb-sid;
  }
  import ieee802-dot1cb-stream-identification-types {
    prefix dot1cb-sid-types;
  }
  import ieee802-dot1cb-frer-types {
    prefix dot1cb-frer-types;
  }
  import ieee802-dot1q-types {
    prefix dot1qtypes;
  }
  import ietf-interfaces {
    prefix if;
  }
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    WG-EMail: stds-802-1-l@ieee.org

    Contact: IEEE 802.1 Working Group Chair
    Postal: C/O IEEE 802.1 Working Group
            IEEE Standards Association
            445 Hoes Lane
            Piscataway, NJ 08854
            USA

    E-mail: stds-802-1-chairs@ieee.org";
  description
    "Management objects that control the frame replication and
    elimination from IEEE Std 802.1CB-2017. This YANG data model conforms
    to the Network Management Datastore Architecture defined in RFC 8342.
    Copyright (C) IEEE (2021). This version of this YANG module is part
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```
    of IEEE Std 802.1CBcv; see the standard itself for full legal
    notices.";
revision 2025-11-12 {
    description
      "Published as part of IEEE Std 802.1CB-2017/Cor 1-2025.";
    reference
      "IEEE Std 802.1CB-2017/Cor 1-2025, Frame Replication and Elimination for
      Reliability - Corrigendum 1.";
  }
revision 2021-12-08 {
    description
      "Published as part of IEEE Std 802.1CBcv-2021. Initial version.";
    reference
      "IEEE Std 802.1CBcv-2021, Frame Replication and Elimination for
      Reliability - FRER YANG Data Model and Management Information Base
      Module.";
  }
  feature auto-configuration {
    description
      "Autoconfiguration of entries in the tables for Stream identity
      table, sequence recovery table, and sequence identification table.";
    reference
      "7.11 of IEEE Std 802.1CB-2017
      10.7 of IEEE Std 802.1CB-2017";
  }
  typedef lan-path-id {
    type int8;
    description
      "An integer specifying a path or LAN. If and only if a packet
      matches an entry in the Sequence identification table that
      specifies HSR or PRP in its frerSeqEncEncapsType object,
      tsnStreamIdLanPathId specifies the LanId or PathId value that must
      be matched for this tsnStreamIdEntry to apply. A value of -1
      indicates that the LanId or PathId are to be ignored.";
    reference
      "10.2.2 of IEEE Std 802.1CB-2017";
  }
  typedef sequence-history-length {
    type uint32 {
      range "2..max";
    }
    description
      "An integer specifying how many bits of the SequenceHistory
      variable are to be used.";
    reference
      "7.4.3.2.2 of IEEE Std 802.1CB-2017";
  }
  grouping sequence-encode-decode {
    description
      "The sequence-encode-decode grouping indicates the type of
      encapsulation used for this instance of the Sequence encode/decode
      function.";
    reference
      "10.5.1.5 of IEEE Std 802.1CB-2017";
    choice encapsulation {
      description
        "This choice indicates the type of encapsulation used for this
        instance of the Sequence encode/decode function. The
        encapsulation includes an Organizationally Unique Identifier
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        (OUI) or Company ID (CID) to identify the organization defining
        the Sequence encode/decode method.";
      reference
        "10.5.1.5 of IEEE Std 802.1CB-2017";
      container r-tag {
        presence "true";
        description
          "R-TAG";
        reference
          "7.8 of IEEE Std 802.1CB-2017";
        leaf type-number {
          type dot1cb-frer-types:seq-encaps-method;
          default "r-tag";
          config false;
          description
            "The type number used for the R-TAG encode/decode method.";
          reference
            "10.5.1.5 of IEEE Std 802.1CB-2017";
        }
        leaf oui-cid {
          type string {
            pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
          }
          default "00-80-C2";
          config false;
          description
            "The Organizationally Unique Identifier (OUI) or Company ID
            (CID) to identify the organization defining the encode/decode
            method. For encode/decode methods defined in IEEE Std
            802.1CB-2017 the OUI/CID is always 00-80-C2.";
          reference
            "10.5.1.5 of IEEE Std 802.1CB-2017";
        }
      }
      container hsr-sequence-tag {
        presence "true";
        description
          "HSR sequence tag";
        reference
          "7.9 of IEEE Std 802.1CB-2017";
        leaf type-number {
          type dot1cb-frer-types:seq-encaps-method;
          default "hsr-seq-tag";
          config false;
          description
            "The type number used for the HSR sequence tag encode/decode
            method.";
          reference
            "10.5.1.5 of IEEE Std 802.1CB-2017";
        }
        leaf oui-cid {
          type string {
            pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
          }
          default "00-80-C2";
          config false;
          description
            "The Organizationally Unique Identifier (OUI) or Company ID
            (CID) to identify the organization defining the encode/decode
```

```
            method. For encode/decode methods defined in IEEE Std
            802.1CB-2017 the OUI/CID is always 00-80-C2.";
          reference
            "10.5.1.5 of IEEE Std 802.1CB-2017";
        }
      }
      container prp-sequence-tag {
        presence "true";
        description
          "PRP sequence trailer";
        reference
          "7.10 of IEEE Std 802.1CB-2017";
        leaf type-number {
          type dot1cb-frer-types:seq-encaps-method;
          default "prp-seq-trailer";
          config false;
          description
            "The type number used for the PRP sequence trailer
            encode/decode method.";
          reference
            "10.5.1.5 of IEEE Std 802.1CB-2017";
        }
        leaf oui-cid {
          type string {
            pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
          }
          default "00-80-C2";
          config false;
          description
            "The Organizationally Unique Identifier (OUI) or Company ID
            (CID) to identify the organization defining the encode/decode
            method. For encode/decode methods defined in IEEE Std
            802.1CB-2017 the OUI/CID is always 00-80-C2.";
          reference
            "10.5.1.5 of IEEE Std 802.1CB-2017";
        }
      }
      container organization-specific {
        presence "true";
        description
          "This container allows to select Sequence encode/decode types
          that are defined by entities outside of IEEE 802.1.";
        reference
          "10.5.1.5 of IEEE Std 802.1CB-2017";
        leaf type-number {
          type int32 {
            range "256..max";
          }
          description
            "The type number used for an encode/decode method defined by
            an entity owning the OUI or CID for this encapsulation type.";
          reference
            "10.5.1.5 of IEEE Std 802.1CB-2017";
        }
        leaf oui-cid {
          type string {
            pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
          }
          description
```

```
          "The Organizationally Unique Identifier (OUI) or Company ID
          (CID) to identify the organization defining the encode/decode
          method.";
        reference
          "10.5.1.5 of IEEE Std 802.1CB-2017";
      }
    }
  }
}
grouping sequence-recovery-algorithm {
  description
    "The sequence-recovery-algorithm grouping specifies which sequence
    recovery algorithm is to be used for this instance of the Sequence
    recovery function.";
  reference
    "10.4.1.5 of IEEE Std 802.1CB-2017";
  choice algorithm {
    description
      "This choice indicates the sequence recovery algorithm used for
      this instance of the Sequence recovery function. It includes an
      Organizationally Unique Identifier (OUI) or Company ID (CID) to
      identify the organization defining the sequence recovery
      algorithm.";
    reference
      "10.4.1.5 of IEEE Std 802.1CB-2017";
    container vector {
      presence "true";
      description
        "Vector Recovery Algorithm.";
      reference
        "10.4.1.5 of IEEE Std 802.1CB-2017";
      leaf type-number {
        type dot1cb-frer-types:seq-rcvy-algorithm;
        default "vector";
        config false;
        description
          "The type number used for the VectorRecoveryAlgorithm.";
        reference
          "10.4.1.5 of IEEE Std 802.1CB-2017";
      }
      leaf oui-cid {
        type string {
          pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
        }
        default "00-80-C2";
        config false;
        description
          "The Organizationally Unique Identifier (OUI) or Company ID
          (CID) to identify the organization defining the sequence
          recovery algorithm. For sequence recovery algorithms defined
          in IEEE Std 802.1CB-2017 the OUI/CID is always 00-80-C2.";
        reference
          "10.4.1.5 of IEEE Std 802.1CB-2017";
      }
    }
    container match {
      presence "true";
      description
        "Match Recovery Algorithm.";
```

```
        reference
          "10.4.1.5 of IEEE Std 802.1CB-2017";
        leaf type-number {
          type dot1cb-frer-types:seq-rcvy-algorithm;
          default "match";
          config false;
          description
            "The type number used for the MatchRecoveryAlgorithm.";
          reference
            "10.4.1.5 of IEEE Std 802.1CB-2017";
        }
        leaf oui-cid {
          type string {
            pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
          }
          default "00-80-C2";
          config false;
          description
            "The Organizationally Unique Identifier (OUI) or Company ID
            (CID) to identify the organization defining the sequence
            recovery algorithm. For sequence recovery algorithms defined
            in IEEE Std 802.1CB-2017 the OUI/CID is always 00-80-C2.";
          reference
            "10.4.1.5 of IEEE Std 802.1CB-2017";
        }
      }
      container organization-specific {
        presence "true";
        description
          "This container allows to select sequence recovery algorithms
          that are defined by entities outside of IEEE 802.1.";
        reference
          "10.4.1.5 of IEEE Std 802.1CB-2017";
        leaf type-number {
          type int32 {
            range "256..max";
          }
          description
            "The type number used for a sequence recovery algorithm
            defined by an entity owning the OUI or CID for this algorithm
            type.";
          reference
            "10.4.1.5 of IEEE Std 802.1CB-2017";
        }
        leaf oui-cid {
          type string {
            pattern "[0-9A-F]{2}(-[0-9A-F]{2}){2}";
          }
          description
            "The Organizationally Unique Identifier (OUI) or Company ID
            (CID) to identify the organization defining the sequence
            recovery algorithm.";
          reference
            "10.4.1.5 of IEEE Std 802.1CB-2017";
        }
      }
    }
  }
  container frer {
```

```
description
  "The managed objects that control Stream identification are
  described in Clause 9 of IEEE Std 802.1CB-2017. The managed objects
  that control FRER are described in Clause 10 of IEEE Std
  802.1CB-2017 as follows:

   a) General requirements on the behavior of counters are in 10.1.
   b) The various tables of managed objects that can manage, in
      detail, each individual Stream, are described in five
      subclauses, including:
     1) Additions to the Stream identity table required for
        Autoconfiguration.
     2) The Sequence generation table that configures instances of
        the Sequence generation function;
     3) The Sequence recovery table that configures instances of the
        Individual recovery function, the Sequence recovery function,
        and the Latent error detection function;
     4) The Sequence identification table that configures instances
        of the Sequence encode/decode function; and
     5) The Stream split table that configures instances of the
        Stream splitting function.
   c) The managed objects that support the automatic configuration,
      upon receipt of a packet, of entries in the first four of the
      preceding tables, are described in the subclause on
      Autoconfiguration.
   d) The per-port, per-Stream packet counters that are kept by FRER
      functions for inspection by network management entities are
      described in 10.8, and the per-port (totaled over all Streams)
      counters in 10.9.

  The managed objects in the subclauses under 9.1 make it possible to
  configure more than one encapsulation for the same stream_handle
  subparameter on the same port. Similarly, the managed objects in
  the subclauses under 10.3 and 10.4 make it possible to configure
  more than one Sequence encode/decode function or more than one
  Sequence generation function for the same stream_handle
  subparameter. [The same value of stream_handle can be in the
  frerSeqGenStreamList of more than one frerSeqGenEntry or in the
  frerSeqRcvyStreamList of more than one frerSeqRcvyEntry.] A system
  shall return an error if an attempt is made to configure
  conflicting requirements upon that system.";
reference
  "Clause 10 of IEEE Std 802.1CB-2017";
list sequence-generation {
  key "index";
  description
    "There is one Sequence generation table in a system, and one
    entry in the Sequence generation table for each Sequence
    generation function. Each frerSeqGenEntry lists the Streams and
    direction for which a single instance of the Sequence generation
    function is to be placed.";
  reference
    "10.3 of IEEE Std 802.1CB-2017";
  leaf index {
    type uint32;
    description
      "Clause 10 of IEEE Std 802.1CB-2017 states that the same Stream
      handle can be present multiple times in the sequence generation
      table. Therefore this index leaf is being used to uniquely
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```
          identify an entry in the sequence generation table.";
      }
      leaf-list stream {
        type leafref {
          path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
        }
        min-elements 1;
        description
          "A list of stream_handle values, corresponding to the values of
          the tsnStreamIdHandle objects in the Stream identity table, on
          which this instance of the Sequence generation function is to
          operate. The single instance of the Sequence generation
          function created by this frerSeqGenEntry operates every packet
          belonging to this Stream, regardless of the port on which it is
          received.";
        reference
          "10.3.1.1 of IEEE Std 802.1CB-2017";
      }
      leaf direction-out-facing {
        type dot1cb-sid-types:direction;
        description
          "An object indicating whether the Sequence generation function
          is to be placed on the out-facing (True) or in-facing (False)
          side of the port.";
        reference
          "10.3.1.2 of IEEE Std 802.1CB-2017";
      }
      leaf reset {
        type boolean;
        description
          "A Boolean object indicating that the Sequence generation
          function is to be reset by calling its corresponding
          SequenceGenerationReset function. Writing the value True to
          frerSeqGenReset triggers a reset; writing the value False has
          no effect. When read, frerSeqGenReset always returns the value
          False.";
        reference
          "10.3.1.3 of IEEE Std 802.1CB-2017";
      }
    }
    list sequence-recovery {
      key "index";
      description
        "There is one Sequence recovery table in a system, and one entry
        in the Sequence recovery table for each Sequence recovery
        function or Individual recovery function that can also be
        present. The entry describes a set of managed objects for the
        single instance of a Base recovery function and Latent error
        detection function included in the Sequence recovery function or
        Individual recovery function. Each frerSeqRcvyEntry lists the
        Streams, ports, and direction for which instances of a Sequence
        recovery function or Individual recovery function are to be
        instantiated.";
      reference
        "10.4 of IEEE Std 802.1CB-2017";
      leaf index {
        type uint32;
        description
          "Clause 10 of IEEE Std 802.1CB-2017 states that the same Stream
```

```
          handle can be present multiple times in the sequence recovery
          table. Therefore this index leaf is being used to uniquely
          identify an entry in the sequence recovery table.";
      }
      leaf-list stream {
        type leafref {
          path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
        }
        min-elements 1;
        description
          "A list of the stream_handle values, corresponding to the
          values of the tsnStreamIdHandle objects in the Stream identity
          table, to which the system is to apply the instance of the
          Sequence recovery function or Individual recovery function.";
        reference
          "10.4.1.1 of IEEE Std 802.1CB-2017";
      }
      leaf-list port {
        type if:interface-ref;
        min-elements 1;
        description
          "The list of ports on each of which the system is to
          instantiate the Sequence recovery function, or from which
          received packets are to be fed to a single instance of the
          Individual recovery function.";
        reference
          "10.4.1.2 of IEEE Std 802.1CB-2017";
      }
      leaf direction-out-facing {
        type dot1cb-sid-types:direction;
        description
          "An object indicating whether the Sequence recovery function or
          Individual recovery function is to be placed on the out-facing
          (True) or in-facing (False) side of the port.";
        reference
          "10.4.1.3 of IEEE Std 802.1CB-2017";
      }
      leaf reset {
        type boolean;
        default "false";
        description
          "A Boolean object indicating that the Sequence recovery
          function or Individual recovery function is to be reset by
          calling its corresponding SequenceGenerationReset function.
          Writing the value True to frerSeqRcvyReset triggers a reset;
          writing the value False has no effect. When read,
          frerSeqRcvyReset always returns the value False.";
        reference
          "10.4.1.4 of IEEE Std 802.1CB-2017";
      }
      container algorithm {
        description
          "This object is an enumerated value specifying which sequence
          recovery algorithm is to be used for this instance of the
          Sequence recovery function. The enumeration uses an OUI or CID
          as shown in Table 10-1. The default value for
          frerSeqRcvyAlgorithm is Vector_Alg (00-80-C2, 0).";
        reference
          "10.4.1.5 of IEEE Std 802.1CB-2017";
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        uses sequence-recovery-algorithm;
      }
      leaf history-length {
        type sequence-history-length;
        default "2";
        description
          "An integer specifying how many bits of the SequenceHistory
          variable are to be used. The minimum and the default value is
          2, maximum is the maximum allowed by the implementation. [Not
          used if frerSeqRcvyAlgorithm = Match_Alg (00-80-C2, 1).]";
        reference
          "10.4.1.6 of IEEE Std 802.1CB-2017";
      }
      leaf reset-timeout {
        type uint32;
        units "ms";
        description
          "An unsigned integer specifying the timeout period in
          milliseconds for the RECOVERY_TIMEOUT event.";
        reference
          "10.4.1.7 of IEEE Std 802.1CB-2017";
      }
      leaf invalid-sequence-value {
        type uint32;
        config false;
        description
          "A read-only unsigned integer value that cannot be encoded in a
          packet as a value for the sequence_number subparameter, i.e.,
          frerSeqRcvyInvalidSequenceValue is larger than or equal to
          RecovSeqSpace.";
        reference
          "10.4.1.8 of IEEE Std 802.1CB-2017";
      }
      leaf take-no-sequence {
        type boolean;
        default "false";
        description
          "A Boolean value specifying whether packets with no
          sequence_number subparameter are to be accepted (True) or not
          (False). Default value False.";
        reference
          "10.4.1.9 of IEEE Std 802.1CB-2017";
      }
      leaf individual-recovery {
        type boolean;
        description
          "A Boolean value specifying whether this entry describes a
          Sequence recovery function or Individual recovery function.
           a) True: The entry describes an Individual recovery function.
              Packets discarded by the SequenceGenerationAlgorithm will
              cause the variable RemainingTicks to be reset. There is no
              Latent error detection function associated with this entry,
              so frerSeqRcvyLatentErrorDetection cannot also be True.
           b) False: The entry describes a Sequence recovery function.
              Packets discarded by the SequenceGenerationAlgorithm will
              not cause the variable RemainingTicks to be reset.";
        reference
          "10.4.1.10 of IEEE Std 802.1CB-2017";
      }
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leaf latent-error-detection {
  type boolean;
  description
    "A Boolean value indicating whether an instance of the Latent
    error detection function is to be instantiated along with the
    Base recovery function in this Sequence recovery function or
    Individual recovery function. frerSeqRcvyLatentErrorDetection
    cannot be set True if frerSeqRcvyIndividualRecovery is also
    True; an Individual recovery function does not include a Latent
    error detection function.";
  reference
    "10.4.1.11 of IEEE Std 802.1CB-2017";
}
container latent-error-detection-parameters {
  description
    "The objects in the following subclauses are present if and
    only if frerSeqRcvyIndividualRecovery is False.";
  reference
    "10.4.1.12 of IEEE Std 802.1CB-2017";
  leaf difference {
    type int32;
    description
      "An integer specifying the maximum difference between
      frerCpsSeqRcvyDiscardedPackets, and the product of
      frerCpsSeqRcvyPassedPackets and (frerSeqRcvyLatentErrorPaths
      - 1) that is allowed. Any larger difference will trigger the
      detection of a latent error by the LatentErrorTest function.";
    reference
      "10.4.1.12.1 of IEEE Std 802.1CB-2017";
  }
  leaf period {
    type uint32;
    units "ms";
    default "2000";
    description
      "The integer number of milliseconds that are to elapse
      between instances of running the LatentErrorTest function. An
      implementation can have a minimum value for
      frerSeqRcvyLatentErrorPeriod, below which it cannot be set,
      but this minimum shall be no larger than 1000 ms (1 s).
      Default value 2000 (2 s).";
    reference
      "10.4.1.12.2 of IEEE Std 802.1CB-2017";
  }
  leaf paths {
    type uint16;
    description
      "The integer number of paths over which FRER is operating for
      this instance of the Base recovery function and Latent error
      detection function.";
    reference
      "10.4.1.12.3 of IEEE Std 802.1CB-2017";
  }
  leaf reset-period {
    type uint32;
    units "ms";
    default "30000";
    description
      "The integer number of milliseconds that are to elapse
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          between instances of running the LatentErrorReset function.
          An implementation can have a minimum value for
          LatentErrorReset, below which it cannot be set, but this
          minimum shall be no larger than 1000 ms (1 s). Default value
          30000 (30 s).";
        reference
          "10.4.1.12.4 of IEEE Std 802.1CB-2017";
      }
    }
  }
  list sequence-identification {
    key "port direction-out-facing";
    description
      "There is one Sequence identification table per system, and one
      entry in the Sequence identification table for each port and
      direction for which an instance of the Sequence encode/decode
      function is to be created. Each entry in the Sequence
      identification table specifies a port and direction on which an
      instance of the Sequence encode/decode function is to be
      instantiated for a list of Streams.";
    reference
      "10.5 of IEEE Std 802.1CB-2017";
    leaf-list stream {
      type leafref {
        path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
      }
      min-elements 1;
      description
        "A list of stream_handles, corresponding to the values of the
        tsnStreamIdHandle objects in the Stream identity table, for
        which the system is to use the same encapsulation for the
        Sequence encode/decode function.";
      reference
        "10.5.1.1 of IEEE Std 802.1CB-2017";
    }
    leaf port {
      type if:interface-ref;
      description
        "The port on which the system is to place an instance of the
        Sequence encode/decode function.";
      reference
        "10.5.1.2 of IEEE Std 802.1CB-2017";
    }
    leaf direction-out-facing {
      type dot1cb-sid-types:direction;
      description
        "An object indicating whether the Sequence encode/decode
        function is to be placed on the out-facing (True) or in-facing
        (False) side of the port.";
      reference
        "10.5.1.3 of IEEE Std 802.1CB-2017";
    }
    leaf active {
      type boolean;
      description
        "A Boolean value specifying whether this frerSeqEncEntry is
        passive (False), and therefore is used only to decode (extract
        information from) input packets passing up the protocol stack,
        or active (True), and therefore is used both for recognizing
```

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          input packets and for encoding output packets being passed down
          the protocol stack.";
        reference
          "10.5.1.4 of IEEE Std 802.1CB-2017";
      }
      container encapsulation {
        description
          "An enumerated value indicating the type of encapsulation used
          for this instance of the Sequence encode/decode function. The
          type includes an OUI or CID.";
        reference
          "10.5.1.5 of IEEE Std 802.1CB-2017";
        uses sequence-encode-decode;
      }
      leaf path-id-lan-id {
        type lan-path-id;
        description
          "A 4-bit integer value to be placed in the PathId field of an
          HSR sequence tag or the LanId field of a PRP sequence trailer
          added to an output packet. This managed object is used only if:
           a) The HSR sequence tag or the PRP sequence trailer is
              selected by the frerSeqEncEncapsType object; and
           b) frerSeqEncActive is False (passive)";
        reference
          "10.5.1.6 of IEEE Std 802.1CB-2017";
      }
    }
    list stream-split {
      key "~~port direction-out-facing~~index";
      description
        "There is one Stream split table per system, with one
        frerSplitEntry per Stream splitting function per set of
        stream_handle values. Each entry in the Stream split table
        specifies a port and direction on which an instance of the Stream
        splitting function is to be instantiated, and the list of
        stream_handles specifying its operation.";
      reference
        "10.6 of IEEE Std 802.1CB-2017";
      leaf index {
        type uint32;
        description
          "Entry in the Stream split table.";
        reference
          "10.6 of IEEE Std 802.1CB-2017";
      }
      leaf port {
        type if:interface-ref;
        description
          "The port on which the system is to place an instance of the
          Stream splitting function performing the stream_handle
          translations specified by frerSplitInputIdList and
          frerSplitOutputIdList is to be placed.";
        reference
          "10.6.1.1 of IEEE Std 802.1CB-2017";
      }
      leaf direction-out-facing {
        type dot1cb-sid-types:direction;
        description
          "An object indicating whether the instance of the Stream
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          splitting function performing the stream_handle translations
          specified by frerSplitInputIdList and frerSplitOutputIdList is
          to be placed on the out-facing (True) or in-facing (False) side
          of the port.";
        reference
          "10.6.1.2 of IEEE Std 802.1CB-2017";
      }
      leaf-list input-id {
        type leafref {
          path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
        }
        min-elements 1;
        description
          "A list of stream_handles (tsnStreamIdHandle values) that are
          to be split.";
        reference
          "10.6.1.3 of IEEE Std 802.1CB-2017";
      }
      leaf-list output-id {
        type leafref {
          path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
        }
        min-elements 1;
        description
          "A list of stream_handles (tsnStreamIdHandle values) into which
          the input packet is to be split, one copy per item in the
          frerSplitOutputIdList.";
        reference
          "10.6.1.4 of IEEE Std 802.1CB-2017";
      }
    }
    container autoconfiguration {
      if-feature "auto-configuration";
      description
        "Container for autoconfiguration managed objects.";
      reference
        "10.7 of IEEE Std 802.1CB-2017";
      list sequence {
        key "index";
        description
          "There is one Sequence autoconfiguration table per system. It
          contains any number of table entries. No two (or more) entries
          in the Sequence autoconfiguration table can have the same
          values for frerAutSeqSeqEncaps, frerAutSeqTagged, and
          frerAutSeqVlan on any given port. Each frerAutSeqEntry object
          relates to a single class of Streams, and specifies how entries
          are created (and destroyed) in the Stream identity table, the
          Sequence recovery table, and the Sequence identification table.";
        reference
          "10.7.1 of IEEE Std 802.1CB-2017";
        leaf index {
          type uint32;
          description
            "Entry in the sequence list referencing to a single class of
            Streams.";
        }
        container sequence-encapsulation {
          description
            "An enumerated value, specifying which Sequence encode/decode
```

```
          function, and therefore, which type sequence_number encoding,
          is to be recognized for the purposes of Autoconfiguration.";
        reference
          "10.7.1.1.1 of IEEE Std 802.1CB-2017";
        uses sequence-encode-decode;
      }
      leaf-list receive-port {
        type if:interface-ref;
        min-elements 1;
        description
          "The list of ports to which this frerAutSeqEntry applies, and
          on which Stream identification functions, Sequence
          encode/decode functions, and Individual recovery functions
          are to be autocreated.";
        reference
          "10.7.1.1.2 of IEEE Std 802.1CB-2017";
      }
      leaf tagged {
        type enumeration {
          enum tagged {
            value 1;
            description
              "A frame must have a VLAN tag to be matched.";
          }
          enum priority {
            value 2;
            description
              "A frame must be untagged, or have a VLAN tag with a VLAN
              ID = 0 to be matched.";
          }
          enum all {
            value 3;
            description
              "A frame is matched whether tagged or not.";
          }
        }
        description
          "An enumerated value indicating whether packets to be matched
          by this frerAutSeqEntry are permitted to have a VLAN tag.";
        reference
          "10.7.1.1.3 of IEEE Std 802.1CB-2017";
      }
      leaf-list vlan {
        type dot1qtypes:vlanid;
        description
          "A list of vlan_identifiers for the packet to match. A null
          list matches all vlan_identifiers.";
        reference
          "10.7.1.1.4 of IEEE Std 802.1CB-2017";
      }
      leaf-list recovery-port {
        type if:interface-ref;
        min-elements 1;
        description
          "The list of ports on which Sequence recovery functions are
          to be autocreated by this frerAutSeqEntry.";
        reference
          "10.7.1.1.5 of IEEE Std 802.1CB-2017";
      }
```

```
        leaf destruction-interval {
          type uint64;
          units "ms";
          default "86400000";
          description
            "An integer number of milliseconds after which an idle set of
            functions created by this frerAutSeqEntry can be destroyed. A
            value of 0 indicates that idle autoconfigured functions are
            not to be destroyed. Default value is 86 400 000 decimal (one
            day).";
          reference
            "10.7.1.1.6 of IEEE Std 802.1CB-2017";
        }
        leaf reset-interval {
          type uint64;
          units "ms";
          description
            "The value used to fill frerSeqRcvyResetMSec when
            autoconfiguring entries in the Sequence recovery table.";
          reference
            "10.7.1.1.7 of IEEE Std 802.1CB-2017";
        }
        container algorithm {
          description
            "The value used to fill frerSeqRcvyAlgorithm when
            autoconfiguring entries in the Sequence recovery table.";
          reference
            "10.7.1.1.8 of IEEE Std 802.1CB-2017";
          uses sequence-recovery-algorithm;
        }
        leaf history-length {
          type sequence-history-length;
          default "2";
          description
            "The value used to fill frerSeqRcvyHistoryLength when
            autoconfiguring entries in the Sequence recovery table.";
          reference
            "10.7.1.1.9 of IEEE Std 802.1CB-2017";
        }
        leaf create-individual {
          type boolean;
          description
            "A Boolean value. If True, the receipt of a packet that
            triggers the autoconfiguration of a new tsnStreamIdEntry also
            triggers the instantiation of a frerSeqRcvyEntry for an
            Individual recovery function.";
          reference
            "10.7.1.1.10 of IEEE Std 802.1CB-2017";
        }
        leaf create-recovery {
          type boolean;
          description
            "A Boolean value. If True, the receipt of a packet that
            triggers the autoconfiguration of a new tsnStreamIdEntry can
            also trigger the instantiation of a frerSeqRcvyEntry for a
            Sequence recovery function.";
          reference
            "10.7.1.1.11 of IEEE Std 802.1CB-2017";
        }
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        leaf latent-error-detection {
          type boolean;
          description
            "A Boolean value. If True, the autoconfiguration of a new
            Sequence recovery function also creates an associated Latent
            Error Detection function.";
          reference
            "10.7.1.1.12 of IEEE Std 802.1CB-2017";
        }
        leaf latent-error-difference {
          type int32;
          description
            "The value used to fill frerSeqRcvyLatentErrorDifference when
            autoconfiguring entries in the Sequence recovery table.";
          reference
            "10.7.1.1.13 of IEEE Std 802.1CB-2017";
        }
        leaf latent-error-period {
          type uint32;
          units "ms";
          default "2000";
          description
            "The value used to fill frerSeqRcvyLatentErrorPeriod when
            autoconfiguring entries in the Sequence recovery table.";
          reference
            "10.7.1.1.14 of IEEE Std 802.1CB-2017";
        }
        leaf latent-error-reset-period {
          type uint32;
          units "ms";
          default "30000";
          description
            "The value used to fill frerSeqRcvyLatentResetPeriod when
            autoconfiguring entries in the Sequence recovery table.";
          reference
            "10.7.1.1.15 of IEEE Std 802.1CB-2017";
        }
      }
      list output {
        key "index";
        description
          "There is one Output autoconfiguration table per system. It
          contains any number of frerAutOutEntry objects, each relating
          to a single class of Streams specifying how active entries are
          created in the Sequence identification table. No two (or more)
          entries in the Output autoconfiguration table can include the
          same port in their frerAutSeqReceivePortList objects.";
        reference
          "10.7.2 of IEEE Std 802.1CB-2017";
        leaf index {
          type uint32;
          description
            "Entry in the output list referencing to a single class of
            Streams.";
        }
        leaf-list port {
          type if:interface-ref;
          min-elements 1;
          description
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          "The list of ports to which this frerAutOutEntry applies, and
          on which active Sequence encode/decode functions are to be
          autocreated.";
        reference
          "10.7.2.1.1 of IEEE Std 802.1CB-2017";
      }
      container encapsulation {
        description
          "An enumerated value, specifying which Sequence encode/decode
          function, and therefore, which type sequence_number encoding,
          is to be used for autoconfigured Streams on the ports in
          frerAutSeqReceivePortList.";
        reference
          "10.7.2.1.2 of IEEE Std 802.1CB-2017";
        uses sequence-encode-decode;
      }
      leaf lan-path-id {
        type lan-path-id;
        description
          "An integer specifying a path or LAN. If and only if
          frerAutOutEncaps specifies HSR or PRP frerAutOutLanPathId
          specifies the LanId or PathId value to be inserted into the
          HSR sequence tag or PRP sequence trailer of autoconfigured
          packets transmitted on the ports in
          frerAutSeqReceivePortList.";
        reference
          "10.7.2.1.3 of IEEE Std 802.1CB-2017";
      }
    }
  }
}
augment "/dot1cb-sid:stream-identity" {
  if-feature "auto-configuration";
  description
    "Two managed objects augment each tsnStreamIdEntry in the Stream
    identity table when Managed objects for autoconfiguration is
    implemented.";
  reference
    "10.2 of IEEE Std 802.1CB-2017";
  leaf auto-configured {
    type boolean;
    config false;
    description
      "A read-only Boolean value, supplied by the system, specifying
      whether this entry was created explicitly (False) or via the
      Sequence autoconfiguration table.";
    reference
      "10.2.1 of IEEE Std 802.1CB-2017";
  }
  leaf lan-path-id {
    type lan-path-id;
    description
      "An integer specifying a path or LAN. If and only if a packet
      matches an entry in the Sequence identification table that
      specifies HSR or PRP in its frerSeqEncEncapsType object,
      tsnStreamIdLanPathId specifies the LanId or PathId value that
      must be matched for this tsnStreamIdEntry to apply. A value of -1
      indicates that the LanId or PathId are to be ignored.";
    reference
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      "10.2.2 of IEEE Std 802.1CB-2017";
  }
}
augment "/if:interfaces/if:interface/if:statistics" {
  description
    "The following counters are the counters for frame replication and
    elimination for reliability. All counters are unsigned integers. If
    used on links faster than 650 000 000 bits per second, they shall
    be 64 bits in length to ensure against excessively short wrap
    times.

    A Stream identification component shall implement the first two
    counters provided in the stream-identification YANG module per-port
    counters, input-packets and output-packets; the remainder of the
    counters in the frer YANG module module are optional for such a
    system.";
  reference
    "10.8 of IEEE Std 802.1CB-2017
    10.9 of IEEE Std 802.1CB-2017";
  container frer {
    description
      "This container contains the per-port as well as the
      per-port-per-stream counters for frame replication and
      elimination for reliability.";
    reference
      "10.8 of IEEE Std 802.1CB-2017
      10.9 of IEEE Std 802.1CB-2017";
    container per-port-counters {
      config false;
      description
        "Contains the per-port counters for frame replication and
        elimination for reliability. The following counters are
        instantiated for each port on which any of the Stream
        identification function, Sequencing function, or Sequence
        encode/decode function is configured. The counters are indexed
        by port number.";
      leaf rx-passed-pkts {
        type uint64;
        config false;
        description
          "The frerCpSeqRcvyPassedPackets counter is incremented once
          for each packet passed up the stack by the
          VectorRecoveryAlgorithm or MatchRecoveryAlgorithm. Its value
          equals the sum (modulo the size of the counters) of all of
          the frerCpsSeqRcvyPassedPackets counters on this same port.";
        reference
          "10.9.1 of IEEE Std 802.1CB-2017";
      }
      leaf rx-discarded-pkts {
        type uint64;
        config false;
        description
          "The frerCpSeqRcvyDiscardPackets counter is incremented once
          for each packet discarded due to a duplicate sequence number
          or for being a rogue packet by any VectorRecoveryAlgorithm or
          MatchRecoveryAlgorithm on this port. Its value equals the sum
          (modulo the size of the counters) of all of the
          frerCpsSeqRcvyRoguePackets and frerCpsSeqRcvyDiscardedPackets
          counters on this same port.";
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          reference
            "10.9.2 of IEEE Std 802.1CB-2017";
        }
        leaf encode-errored-pkts {
          type uint64;
          config false;
          description
            "The frerCpSeqEncErroredPackets counter is incremented once
            each time the Sequence encode/decode function receives a
            packet that it is unable to decode successfully. Its value
            equals the sum (modulo the size of the counters) of all of
            the frerCpsSeqEncErroredPackets counters on this same port.";
          reference
            "10.9.2 of IEEE Std 802.1CB-2017";
        }
      }
      list per-port-per-stream-counters {
        key "direction-out-facing handle";
        config false;
        description
          "Contains the per-port-per-stream counters for frame
          replication and elimination for reliability. The following
          counters are instantiated for each port on which any of the
          Stream identification function, Sequencing function, or
          Sequence encode/decode function is configured. The counters are
          indexed by port number, facing (in-facing or out-facing), and
          stream_handle value.";
        reference
          "10.8 of IEEE Std 802.1CB-2017";
        leaf direction-out-facing {
          type dot1cb-sid-types:direction;
          description
            "An object indicating whether the counters apply to
            out-facing (True) or in-facing (False).";
        }
        leaf handle {
          type leafref {
            path '/dot1cb-sid:stream-identity/dot1cb-sid:handle';
          }
          description
            "The according tsnStreamIdHandle for the counters.";
        }
        leaf generation-reset {
          type uint64;
          config false;
          description
            "The frerCpsSeqGenResets counter is incremented each time the
            SequenceGenerationReset function is called.";
          reference
            "10.8.2 of IEEE Std 802.1CB-2017";
        }
        leaf rx-out-of-order-pkts {
          type uint64;
          config false;
          description
            "The frerCpsSeqRcvyOutOfOrderPackets counter is incremented
            once for each packet accepted out-of-order by the
            VectorRecoveryAlgorithm or MatchRecoveryAlgorithm.
            Out-of-order means that the packet's sequence number is not
```

```
            one more than the previous packet received.";
          reference
            "10.8.3 of IEEE Std 802.1CB-2017";
        }
        leaf rx-rogue-pkts {
          type uint64;
          config false;
          description
            "The frerCpsSeqRcvyRoguePackets counter is incremented once
            for each packet discarded by the VectorRecoveryAlgorithm
            because its sequence_number subparameter is more than
            frerSeqRcvyHistoryLength from RecovSeqNum.";
          reference
            "10.8.4 of IEEE Std 802.1CB-2017";
        }
        leaf rx-passed-pkts {
          type uint64;
          config false;
          description
            "The frerCpsSeqRcvyPassedPackets counter is incremented once
            for each packet passed up the stack by the
            VectorRecoveryAlgorithm or MatchRecoveryAlgorithm.";
          reference
            "10.8.5 of IEEE Std 802.1CB-2017";
        }
        leaf rx-discarded-pkts {
          type uint64;
          config false;
          description
            "The frerCpsSeqRcvyDiscardedPackets counter is incremented
            once for each packet discarded due to a duplicate sequence
            number by the VectorRecoveryAlgorithm or
            MatchRecoveryAlgorithm.";
          reference
            "10.8.6 of IEEE Std 802.1CB-2017";
        }
        leaf rx-lost-pkts {
          type uint64;
          config false;
          description
            "The frerCpsSeqRcvyLostPackets counter is incremented once
            for each packet lost by the VectorRecoveryAlgorithm. A packet
            is counted as lost if its sequence number is not received on
            any ingress port.

            NOTE—If per-source sequence numbering is used,
            frerCpsSeqRcvyLostPackets can count, as lost, packets that
            were sent to another destination, but not lost.";
          reference
            "10.8.7 of IEEE Std 802.1CB-2017";
        }
        leaf rx-tagless-pkts {
          type uint64;
          config false;
          description
            "The frerCpsSeqRcvyTaglessPackets counter is incremented once
            for each packet received by the VectorRecoveryAlgorithm that
            has no sequence_number subparameter.";
          reference
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          "10.8.8 of IEEE Std 802.1CB-2017";
      }
      leaf rx-resets {
        type uint64;
        config false;
        description
          "The frerCpsSeqRcvyResets counter is incremented once each
          time the SequenceRecoveryReset function is called.";
        reference
          "10.8.9 of IEEE Std 802.1CB-2017";
      }
      leaf rx-latent-error-resets {
        type uint64;
        config false;
        description
          "The frerCpsSeqRcvyLatentErrorResets counter is incremented
          once each time the LatentErrorReset function is called.";
        reference
          "10.8.10 of IEEE Std 802.1CB-2017";
      }
      leaf encode-errored-pkts {
        type uint64;
        config false;
        description
          "The frerCpsSeqEncErroredPackets counter is incremented once
          each time the Sequence encode/decode function receives a
          packet that it is unable to decode successfully.";
        reference
          "10.8.11 of IEEE Std 802.1CB-2017";
      }
    }
  }
 }
}
```